FIVE-POINT CHARACTERIZATION OF CLOSED IMMERSIONS OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2905-2918 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 2920-2927, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The chapter lemma equating universally injective, monomorphism and unramified conditions (lemma-universally-injective-unramified, same chapter) and the schemes characterization (Etale Morphisms, characterize-closed-immersion); cited inputs with the representability reductions written in the retained span.

Source reference. Original lemma 04XV: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2927–2927. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05W6: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2867–2877. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. The following are equivalent:*

1. *f is a closed immersion,*
2. *f is universally closed, unramified, and a monomorphism,*
3. *f is universally closed, unramified, and universally injective,*
4. *f is universally closed, locally of finite type, and a monomorphism,*
5. *f is universally closed, universally injective, locally of finite type, and formally unramified.*

Proof. The equivalence of (2) – (5) follows immediately from Lemma 05W6. Moreover, if (2) – (5) are satisfied then f is representable. Similarly, if (1) is satisfied then f is representable. Hence the result follows from the case of schemes, see Étale Morphisms, Lemma 04XV. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.